

S2F-agent: Skill-grounded agent for Sequence-to-Function computational genomics workflows

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Abstract

Sequence-to-Function (S2F) foundation models are revolutionizing genomic research, yet their fragmented ecosystem severely bottlenecks practical application by incompatible inputs, outputs, and runtime environments. General-purpose coding agents lack the strict domain constraints necessary to resolve these biological intricacies safely. Here, we present s2f-agent, a skill-grounded agent orchestration system that translates open-ended genomics queries into reproducible, executable analysis. By integrating canonical input keys, task-specific playbooks, and normalized contracts, s2f-agent unifies workflows across 11 state-of-the-art models, including AlphaGenome, Borzoi, and Evo 2. Validated through rigorous routing and groundedness evaluations, s2f-agent bridges the critical gap between complex model architectures and practical utility, effectively transforming an unwieldy ecosystem into an accessible operational layer for researchers.

Introduction

A central goal of modern genomics is to explain how DNA sequence gives rise to molecular function, especially within the non-coding genome where many trait- and disease-associated variants reside¹. Sequence-to-function (S2F) models were developed to address this problem by learning from sequence signals that are informative about downstream biology². In a broad operational sense, S2F models include both direct predictive models that map sequence to molecular phenotypes and genomic language or foundation models (gLMs) that learn reusable functional representations from large unlabeled sequence corpora and support downstream inference through embeddings, zero-shot prediction, fine-tuning, or generative design³. Under this broader view, the purpose of S2F modeling is not only predictive accuracy in isolation, but also the recovery of usable regulatory hypotheses from sequence alone: which variants are likely functional, which loci may alter gene regulation, and which genomic intervals merit further experimental follow-up.

The significance of this broadened S2F paradigm is increasingly clear. Direct predictive models provide explicit readouts for chromatin accessibility, transcription, splicing, 3D genome features, and variant effects, such as AlphaGenome⁴, Borzoi⁵, and SegmentNT⁶, whereas gLMs expand the field by uncovering regulatory grammar and evolutionary constraints from multispecies or genome-scale unlabeled sequence data, such as DNABERT⁷, Evo⁸, GPN⁹, Nucleotide Transformer¹⁰. As a result, S2F systems now support non-coding variant interpretation, prioritization of candidate causal variants within GWAS loci, promoter and regulatory element analysis, sequence embedding workflows, cross-species functional transfer, and computational guidance for downstream clinical genetics, target discovery, and precision-medicine applications. That diversity is valuable, but it also creates an operational problem: each model family comes with its own assumptions about assemblies, coordinate systems, sequence formatting, runtime dependencies, output heads, context windows, and interpretation boundaries. As a result, even relatively simple requests such as "score this variant," "embed this interval," or "run track prediction on hg38" often require substantial tool-specific translation before they can become reliable analysis steps.

Recent agentic systems have shown that large language models can assist with software development and scientific workflows^{11,12}, but open-ended assistance alone is not enough for genomics. The growing importance and heterogeneity of S2F models actually raises the bar for orchestration: if a system silently infers the wrong assembly, mixes 0-based and 1-based coordinates, ignores output-head requirements, confuses an embedding workflow with a supervised prediction workflow, or invents unsupported API symbols, the resulting analysis can become scientifically misleading rather than merely inconvenient. A useful genomics agent therefore needs more than a conversational interface. It needs explicit routing behavior, domain-grounded constraints, task-aware input validation, and outputs that can be inspected, executed, and regression-tested.

To overcome these operational bottlenecks, we introduce s2f-agent, a skill-grounded orchestration layer that translates open-ended genomics queries into executable, normalized analysis plans. Our system elegantly decouples orchestration logic from model-specific intricacies using a layered architecture. By combining intelligent routing policies with canonical input keys and strict task contracts, s2f-agent automatically resolves ambiguous inputs and standardizes expected outputs—ensuring every generated plan includes explicit assumptions, runnable steps, and predefined fallback behaviors. Our work presents three primary contributions: (1) a skill-package abstraction that modularizes genomics model capabilities into

inspectable, routable units; (2) a contract-aware framework that unifies workflow generation across a highly heterogeneous model landscape; and (3) comprehensive evaluation suites that rigorously test routing accuracy, groundedness, and overall plan actionability. Together, these elements provide a reliable blueprint for deploying S2F models without overstating the underlying biological benchmarks.

Results

s2f-agent provides a layered orchestration architecture for genomics

To ensure reliability and inspectability, s2f-agent implements a layered runtime architecture that strictly separates high-level orchestration from model-specific execution (Figure 1). This architecture is organized into six functional tiers: The agent dictates the core mission, routing behavior, and safety boundaries. The registry acts as the machine-readable control surface, governing canonical input handling, task and output contracts, and recovery policies. The skills layer encapsulates model-family-specific operational knowledge (e.g., SKILL.md, skill.yaml, references, and helper scripts), while playbooks standardize reusable, task-level workflows across different skills. The evals layer ensures robustness through curated behavioral and structural checks. Finally, the scripts layer exposes reproducible runtime entry points for every stage of the pipeline—from routing and orchestration to execution and smoke testing.

Crucially, this separation of concerns transforms open-ended user requests from a single opaque prompt into a staged, inspectable pipeline. The workflow initiates with `route_query.sh`, which determines the primary task and optimal lead skill, explicitly triggering a clarifying action if biological inputs are dangerously underspecified. Next, `run_agent.sh` synthesizes the routed skill with strict I/O contracts to generate a normalized plan object. Finally, `execute_plan.sh` dry-runs or executes the runnable steps, verifying the generation of expected downstream artifacts.

This architectural paradigm is particularly critical in complex genetic analyses, where systems that appear flexible but mask implicit biological assumptions can be actively harmful. By treating standardized structure as a core feature rather than a constraint, s2f-agent avoids acting as an autonomous, unconstrained scientific reasoner. Instead, it serves as a transparent orchestration layer, safely translating underspecified intent into executable, rigorously defined workflow contracts.

Unifying the S2F ecosystem via Skills and Playbooks

To provide comprehensive coverage of the broadened S2F landscape, s2f-agent decouples specific model capabilities (Skills) from generalized biological tasks (Playbooks). This dual abstraction ensures that model-specific syntax remains isolated, while a shared orchestration layer coordinates complex workflows.

At the model level, the current registry packages 11 skills, successfully capturing the capabilities of state-of-the-art S2F systems (**Figure 1A**). This includes stable environments for AlphaGenome, Borzoi, DNABERT-2, Evo 2, GPN, Nucleotide Transformer v3, and SegmentNT. By localizing model-specific syntax within the skill layer, the system seamlessly spans predictive models and self-supervised genomic language models used for zero-shot functional inference or downstream sequence design.

Crucially, users rarely query a repository structure; they query biological objectives. Therefore, these skills are orchestrated by Playbooks across four mature workflow classes: Track Prediction Normalizes requirements for species, assembly, interval, and output heads to structure multi-track genomic predictions (**Figure 1B**). Embedding Generation Requires sequence/interval and explicit embedding targets to differentiate between specific representation strategies. Fine-Tuning Aligns training plans with expected artifacts by mandating dataset schemas, task objectives, and compute constraints. Variant-Effect Analysis Enforces contracts for assembly, interval/coordinate, and allele specification before generating execution steps for variant scoring.

By crossing these two layers, s2f-agent functions as a true routing engine. When a user requests variant-effect analysis, the playbook enforces the required biological inputs, and the agent intelligently routes the request to the most appropriate skill (e.g., AlphaGenome, GPN or Borzoi). This architecture proves that s2f-agent is not a rigid wrapper for a single model, but a reusable, cross-model orchestration layer for modern genomics.

Contract-Driven Orchestration and Validation

At the core of s2f-agent is a rigorous contract layer that translates free-form user intent into scientifically safe, testable orchestration (**Figure 1C**). Grounded in a structured input schema, the system normalizes essential biological context from natural language input, while establishing strict task contracts that define the non-negotiable minimum inputs for distinct workflows.

Crucially, this structural strictness changes how the agent behaves when faced with incomplete information (**Figure 1D**). Rather than optimistically injecting implicit assumptions to satisfy ambiguous prompts, s2f-agent clarifies to a safety-first routing decision. By explicitly halting execution to expose missing parameters (e.g., assembly aliases, coordinate conventions, unspecified alleles or undefined output heads), the system prevents biologically invalid analyses under ambiguous conditions. Concurrently, output contracts enforce uniform structural guarantees across all generated plans, including explicit assumptions, reproducible steps, and defined fallbacks, making highly heterogeneous tasks comparable at the orchestration level.

The fidelity of this contract-based architecture is validated by regression-oriented evaluation suites (**Supplementary Figure 1**). Rather than chasing generalized conversational benchmarks or leaderboard supremacy, these evaluations are designed to strictly detect biological drift. A routing suite ensures stable skill invocation and ambiguity handling; a groundedness suite strictly monitors against hallucination-like fabrications or unsupported API behaviors; and a task-success suite verifies that the generated plans are minimally sufficient and actionable. By rigorously testing routing reliability, groundedness against hallucination-like fabrications, and task actionability, these curated evaluations ensure that s2f-agent operates not as an unpredictable generative layer, but as a transparent, reproducible orchestration engine tailored for the strict demands of computational genomics. Together, this integrated contract-and-evaluation paradigm demonstrates that s2f-agent is engineered as a highly testable, reproducible orchestration layer, prioritizing scientific validity over mere conversational fluency.

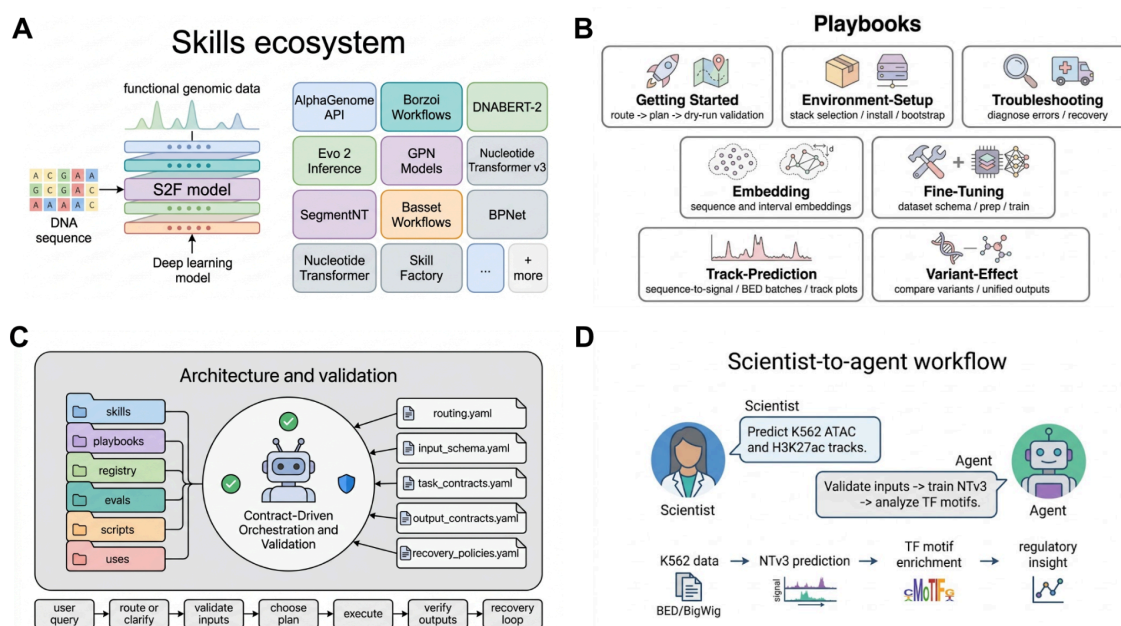


Figure 1. Overview of s2f-agent as a contract-driven agent system for sequence-to-function model workflows.

The figure summarizes the major components and operating logic of s2f-agent, including the Skills ecosystem (A), Playbooks (B), Agent Architecture (C), and Scientist-to-agent workflow (D).

(A) The Skills ecosystem panel illustrates how sequence-to-function models transform DNA sequence inputs into functional genomic outputs, alongside representative modular skills exposed by the system, including AlphaGenome API, Borzoi Workflows, DNABERT-2, Evo 2, GPN, Nucleotide Transformer, SegmentNT, Basset, BPNet, and Skill Factory.

(B) The Playbooks panel shows standardized workflow controllers for common biological tasks, including getting started, environment setup, troubleshooting, embedding, fine-tuning, track prediction, and variant-effect analysis.

(C) The Architecture and validation panel depicts the central agent as a contract-driven orchestrator that connects repository modules, routing logic, input schemas, task and output contracts, and recovery policies. The bottom workflow indicates how user queries are routed or clarified, validated, planned, executed, verified, and recovered when needed.

(D) The Scientist-to-agent workflow panel illustrates a representative scenario in which the scientist asks a task, and the agent validates genomic inputs, selects a sequence-to-function model, runs the analysis pipeline, and returns biological insight.

Decoding regulatory grammar through foundational model fine-tuning on functional profiles

To evaluate the orchestration of fine-tuning workflows on continuous functional genomic profiles, we constructed a real-data benchmark using ENCODE K562 ATAC-seq and H3K27ac ChIP-seq assays mapped to the hg38 assembly. Active regulatory regions were defined by intersecting ATAC and H3K27ac peak sets. We centered fixed 4,096-bp windows on these overlaps and paired them with randomly sampled, non-overlapping negative windows of identical length. This procedure yielded a balanced dataset of 64,198 total windows (32,099 positive, 32,099 negative). To ensure rigorous evaluation, we reserved chromosome 8 as a strict holdout test set and assigned the remaining windows to training and development splits via a deterministic

hash-based partition, yielding 52,395 training, 9,283 development, and 2,520 test windows. For each window, underlying genomic sequences were extracted from HG38 reference and paired with assay signals summarized as the mean fold-change over control from the corresponding ENCODE bigWig tracks. Consequently, each training example paired a genomic sequence with two continuous regression targets: `atac_fc_mean` and `h3k27ac_fc_mean`.

We then evaluated a lightweight adaptation strategy utilizing the Nucleotide Transformer v3 (NTv3). Specifically, the pre-trained InstaDeepAI/NTv3_100M_post backbone was kept frozen, and only a two-output linear regression head was trained on the pooled sequence features. A baseline fine-tuning run was executed across the full training and development splits for 10 epochs using a learning rate of 0.01 and no weight decay. Training loss decreased monotonically from 30.28 to 10.01, indicating that the regression head successfully learned a stable rescaling of the frozen representations (**Figure 2A**). On the full chromosome 8 holdout set, this head-only model achieved a mean Pearson correlation of 0.38 across both targets, performing notably better for H3K27ac (Pearson 0.56) than for ATAC (Pearson 0.69) (**Figure 2B, and Supplementary Figure 2A**). By comparison between native pre-trained outputs and our fine-tuned regression head, we observed that the distribution of prediction rapidly shifted to real K562 ATAC-seq and H3K27ac ChIP-seq signals after fine-tuning (**Figure 2C-D, and Supplementary Figure 2B-C**).

To elucidate the non-linear regulatory relationship between chromatin accessibility (ATAC-seq) and active enhancer/promoter marks (H3K27ac), we systematically analyzed the discordant ATAC-seq and H3K27ac Regions of the K562 cell line. Across the held-out set of 2,520 genomic intervals, we defined and isolated two distinct classes of discordant regulatory regions: ATAC-high/H3K27ac-low (hereafter A-high/H-low) and H3K27ac-high/ATAC-low (H-high/A-low) (**Figure 2F**). Subsequent Position Weight Matrix (PWM) and transcription factor (TF) enrichment analysis of the H-high/A-low regions revealed a highly robust, lineage-specific regulatory signature (**Figure 2G**). The most statistically significant enrichments were dominated by the Krüppel-like factor (KLF) family, including KLF14 ($q=5.76 \times 10^{-11}$), KLF5 ($q=5.16 \times 10^{-10}$), and the critical erythroid lineage master regulator KLF1 ($q=2.21 \times 10^{-9}$). Given the erythroleukemic origin of K562 cells, the overrepresentation of KLF1 strongly indicates that these highly acetylated, yet less accessible domains likely correspond to the nucleosome-dense flanking regions of erythroid-specific super-enhancers. Furthermore, we observed a highly significant enrichment of ZNF143 ($q=3.66 \times 10^{-10}$), a structural factor known to mediate active enhancer-promoter looping, highlighting the 3D topological importance of these domains. Conversely, the A-high/H-low regions exhibited a markedly different transcription factor binding landscape. This subset was uniquely enriched for members of the ETS transcription factor family, specifically ETV5::DRGX ($q=0.00487$) and ETV2::DRGX ($q=0.00768$). Because these genomic loci exhibit high chromatin accessibility but lack the H3K27ac modification requisite for active transcription, our analysis posits that these sites represent "primed" or "poised" enhancers. The binding of ETS family members—often functioning as pioneer factors—suggests they are responsible for initiating chromatin opening, pre-conditioning these loci for subsequent activation events.

Together, these results demonstrate that automated analytical orchestration via s2f-agent effectively disentangles complex, multi-modal epigenomic states, translating continuous sequence-to-function predictions into a highly interpretable, biologically actionable landscape of the transcription factor networks driving cell identity.

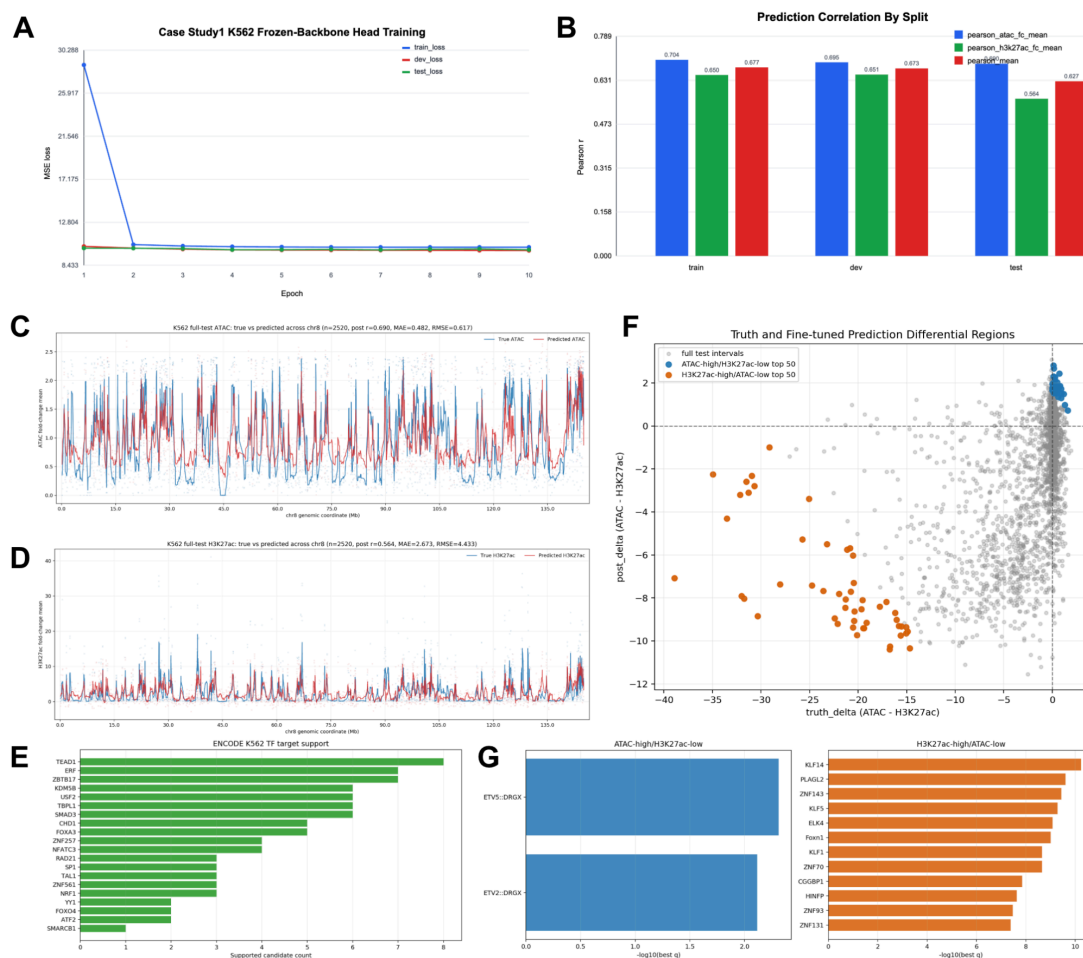


Figure 2. Case study of s2f-agent for decoding regulatory grammar through foundational models.

(A) Training curves for the K562 frozen-backbone, head-only regression model across 10 epochs, showing MSE loss for train, development, and test splits.

(B) Pearson correlation by split for ATAC fold-change mean, H3K27ac fold-change mean, and their average. On the held-out test set, the model achieved Pearson $r = 0.690$ for ATAC, $r = 0.564$ for H3K27ac, and mean $r = 0.627$.

(C-D) Genome-wide track comparison across the full chr8 test set of 2,520 intervals. True and predicted fold-change means are shown as smoothed curves over genomic midpoint coordinates for ATAC (C) and H3K27ac (D). Test-set performance was $r = 0.690$, $MAE = 0.482$ for ATAC and $r = 0.564$, $MAE = 2.673$ for H3K27ac.

(E) ENCODE K562 transcription-factor support among selected top candidate regulatory intervals. Bars indicate the number of candidate intervals supported by ENCODE TF ChIP-seq evidence; the strongest support included KDM5B, ZBTB17, ERF, TBPL1, CHD1, TEAD1, and RAD21.

(F) Differential regulatory regions selected for PWM enrichment analysis. Each point represents a full-test interval, plotted by observed ATAC-H3K27ac delta versus fine-tuned predicted ATAC-H3K27ac delta. Highlighted points mark the top 50 ATAC-high/H3K27ac-low and top 50 H3K27ac-high/ATAC-low regions with concordant observed and predicted delta direction.

(G) Top transcription factors enriched in PWM motif analysis for the two differential region classes. Bar length represents $-\log_{10}(\text{best } q \text{ value})$, summarizing AME enrichment over JASPAR 2026 vertebrate motifs. ATAC-high/H3K27ac-low regions were enriched for ETV-family composite motifs, whereas H3K27ac-high/ATAC-low regions showed strong enrichment for KLF, ZNF, SP, and related regulatory motifs.

Unifying heterogeneous workflows for multi-perspective variant effect inference

To demonstrate the practical utility of s2f-agent in standardizing sequence-to-function (S2F) analysis, we implemented an end-to-end batch variant-effect pipeline. This orchestration layer naturally accommodates inputs via VCF or CSV, routing tasks to targeted execution modules, such as the alphagenome, GPN, and Borzoi, based on request logic. By enforcing strict task contracts (which require explicit assembly, coordinate systems, and allele specifications) alongside structured output contracts, s2f-agent transforms isolated, model-specific scripts into reproducible, machine-checkable execution paths. For example, the variant-effect playbook dynamically handles both SNPs and multibase alleles, transparently propagates VCF INFO fields, and emits wide-format, per-tissue effect summaries. This operational legibility not only guarantees the generation of concrete, verifiable artifacts, but exposes the underlying computational assumptions, making heterogeneous genomics workflows fundamentally more reproducible (**Figure 3A**).

Building upon this standardized framework, we evaluated whether multi-model orchestration could improve functional prioritization in a realistic complex disease setting. We analyzed a curated set of 907 fine-mapped, non-coding variants associated with coronary artery disease (CAD) in hg38 coordinates. Utilizing s2f-agent, we assembled a unified variant-level table integrating three complementary S2F models: AlphaGenome for tissue-specific regulatory effects (extracting the Liver_mean_diff output), Borzoi for transcript-linked contact perturbation (derived from the RNA-SAD mean across tracks), and GPN for evolutionary constraint (based on the mean forward/reverse log-likelihood-ratio) (**Figure 3B**). Because reported GWAS risk alleles can correspond to either the reference or alternate sequence, all raw model outputs were systematically re-oriented to the risk allele using a risk_is_ref flag, ensuring that positive and negative values consistently reflected the same biological directionality. Each oriented score was then standardized across the 907 variants to generate z-tissue, z-contact, and z-evolutionary (**Figure 3C**). The final joint prioritization score, defined as the arithmetic mean of these three z-scores, was used to rank the variants, with the top 30 selected for focused downstream interpretation (**Figure 3D, and Supplementary Figure 3A**).

Externally, the analytical pipeline executed automated queries against GTEx single-tissue eQTL data and the GWAS Catalog. For GTEx analyses, the normalized effect size (nes) was converted into a risk-allele-oriented metric (risk_nes), where positive and negative values indicate risk-allele-associated upregulation and downregulation, respectively. Variants were statistically classified as demonstrating significant expression impact if they possessed at least one GTEx association meeting genome-wide significance ($p < 5 \times 10^{-8}$), and as stronger eQTL candidates if this association also satisfied an absolute effect size threshold of $|\text{risk_nes}| \geq 0.5$. To anchor our CAD interpretations, we specifically tracked peak associations within pathologically relevant tissues, including liver, whole blood, and three arterial beds (**Figure 3E**). Concurrently, GWAS Catalog associations were programmatically retrieved by rsID to determine if the prioritized variants retained independent, pleiotropic support from CAD- or lipid-related phenotypes (**Figure 3F**).

Crucially, this joint prioritization strategy captured a robust set of regulatory candidates that would be difficult to recover using any single model in isolation. Pairwise correlations among the three model-specific scores approached zero, yet each maintained a moderate association with the joint score, demonstrating that the final ranking integrates orthogonal biological signals rather

than redundant predictions. External validation confirmed this functional enrichment: 20 of the top 30 prioritized variants exhibited genome-wide significant eQTL evidence in GTEx, with 10 meeting the more stringent absolute effect-size threshold (**Supplementary Figure 3B**). Both up-regulatory and down-regulatory patterns were observed across target tissues. The most prominent candidates, such as rs12740374 and rs7528419, were strongly corroborated by highly significant eQTLs and independent GWAS hits (**Supplementary Figure 3C**).

Together, these results demonstrate that multi-model orchestration via s2f-agent refines non-coding variant prioritization, effectively filtering out model-specific noise to highlight a compact, biologically interpretable subset of actionable CAD candidates with reproducible external support.

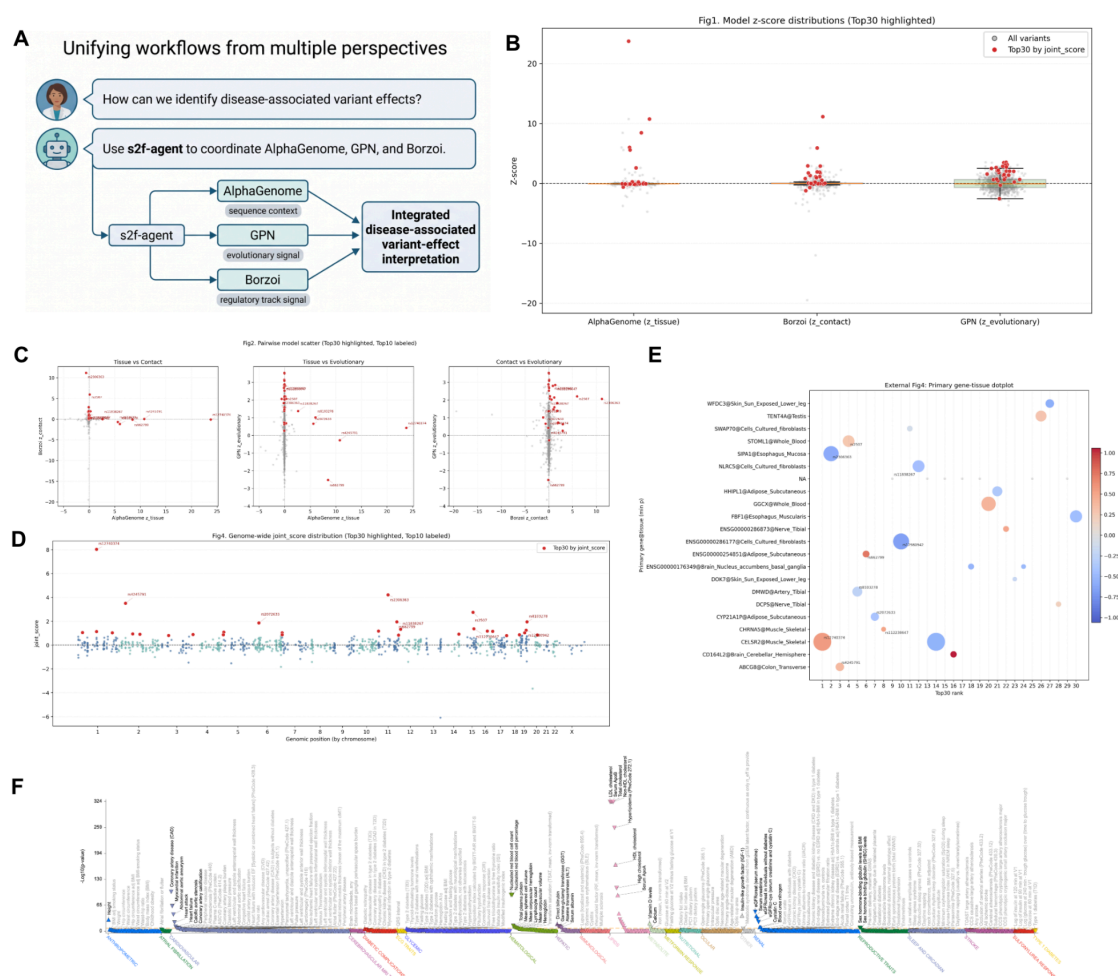


Figure 3. Case study of s2f-agent for quantifying variant effects through multi-perspective models.

(A) The schematic illustrates how a genetic researcher's question is transformed into a contract-driven variant-effect workflow by the s2f-agent. Given disease-associated GWAS variants, the agent validates required inputs such as genome assembly, coordinates, REF/ALT alleles, and VCF records, then selects the variant-effect playbook and orchestrates three complementary skills. AlphaGenome estimates tissue-relevant functional effects, Borzoi evaluates long-range regulatory sequence-to-signal perturbations, and GPN provides evolutionary constraint or allele-likelihood evidence. The heterogeneous model outputs are standardized into a unified table and integrated to prioritize candidate disease-associated variants, demonstrating how s2f-agent unifies multiple model perspectives into a reproducible biological interpretation workflow.

(B) Distribution of risk-allele-oriented model z-scores across 907 CAD fine-mapped noncoding variants. Scores from AlphaGenome (z_{tissue}), Borzoi (z_{contact}), and GPN ($z_{\text{evolutionary}}$) were standardized across variants; red points indicate the Top30 variants ranked by the mean joint score.

(C) Pairwise comparison of the three model-specific z-scores. Top30 variants are highlighted and Top10 variants are labeled, showing that high-priority variants arise from complementary model signals rather than strong pairwise model correlation.

(D) Manhattan-style genome-wide distribution of the joint prioritization score, defined as the arithmetic mean of z_{tissue} , z_{contact} , and $z_{\text{evolutionary}}$. Top30 variants are highlighted and Top10 variants are labeled, with rs12740374 showing the strongest joint score.

(E) External GTEx eQTL validation of the Top30 variants. Each point shows the primary gene-tissue association with the minimum GTEx p value for a prioritized variant. Point size reflects eQTL significance, and color indicates risk-allele-oriented normalized effect size (risk_nes), with positive values indicating risk-allele-associated upregulation and negative values indicating downregulation.

(F) GWAS Catalog/PheWAS-style phenotype association plot for prioritized variants, showing cross-phenotype support among CAD, lipid, metabolic, cardiovascular, and related traits. The dashed line marks the nominal/significance threshold used in the plot, highlighting recurrent disease-relevant associations among the prioritized loci.

Discussion

s2f-agent addresses a practical and increasingly important problem in computational genomics: how to translate open-ended workflow requests into safe, structured, and executable plans across heterogeneous S2F model families. In the broader framing adopted here, that family includes both direct predictive models and gLM-style foundation models. This problem matters because S2F systems are no longer niche benchmark tools. They are increasingly used to interpret non-coding variation, prioritize candidate causal variants, reason about transcriptional and splicing consequences, support embedding- or zero-shot-driven functional inference, and enable downstream translational workflows in clinical genetics and therapeutic discovery. The system does not try to replace AlphaGenome, Borzoi, DNABERT-2, Evo 2, GPN, NTV3, or SegmentNT. Instead, it provides the orchestration layer that sits above them and standardizes how requests are routed, validated, and operationalized.

This framing is important for two reasons. First, many real-world bottlenecks in applying S2F models are not caused by a lack of predictive architectures, but by difficulty choosing among direct predictors versus embedding-oriented gLM workflows, supplying the right inputs, respecting model-specific constraints, and turning user intent into reproducible execution steps. Second, the scientific value of an agent in this setting depends less on rhetorical fluency than on whether it makes hidden assumptions visible and whether its outputs can be validated. s2f-agent was designed with these priorities in mind through explicit contracts, clarification logic, packaged skills, and regression-oriented evaluation.

At the same time, the current system has clear limits. The evaluation layer is curated and case-based rather than population-scale. Skill maturity is uneven, with some workflows stabilized and others still in development. The strongest current evidence lies in representative case studies and normalized plan generation, not in head-to-head scientific benchmarking against alternative agent frameworks. In addition, the quality of downstream execution remains dependent on the correctness, availability, and maintenance status of the underlying model ecosystems and external APIs.

These limits also suggest natural next steps. Future work could add stronger self-critique loops, richer execution verifiers, and optional external knowledge augmentation for track or tissue context. The system could also be extended toward broader benchmarking, larger real-world task corpora, and tighter integration between generated plans and downstream artifact inspection. However, those directions should be understood as extensions to the current architecture rather than results already demonstrated here.

Code availability

<https://github.com/JiaqiLiZju/s2f-agent>

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Author contributions

J.L. collected the data, implemented the model, performed downstream data analyses, and wrote the initial draft of the manuscript.

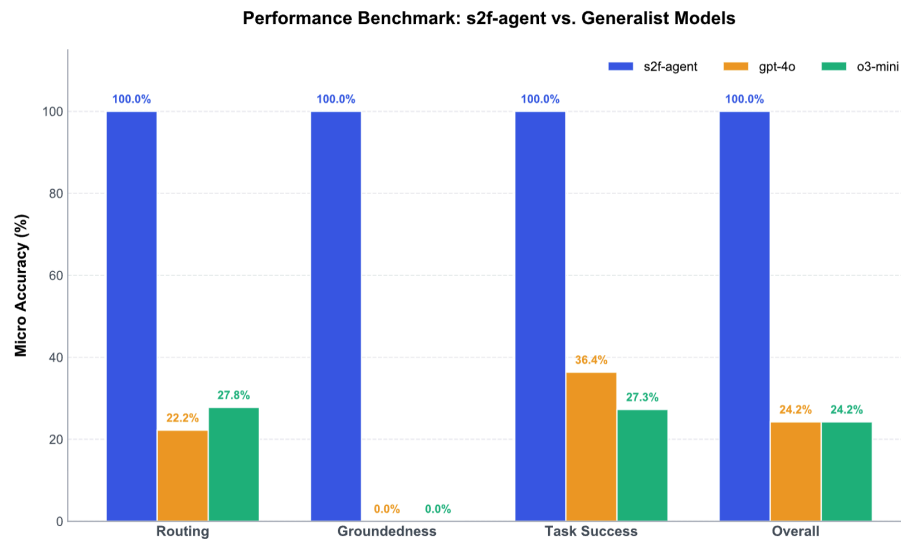
Competing interests

Authors declare that they have no competing interests.

Ethics approval

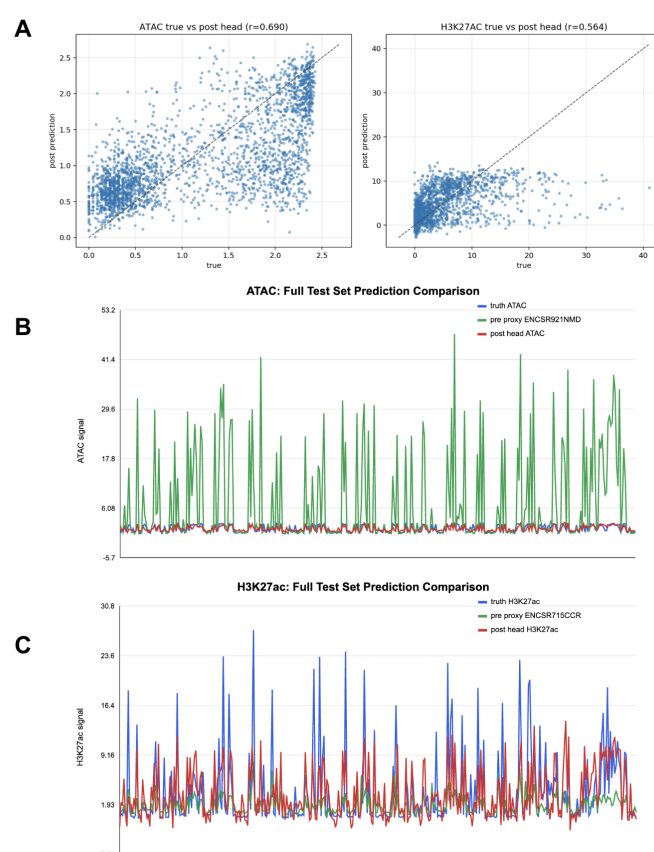
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Supplementary Figures



Supplementary Figure 1. Performance benchmark of s2f-agent against generalist models.

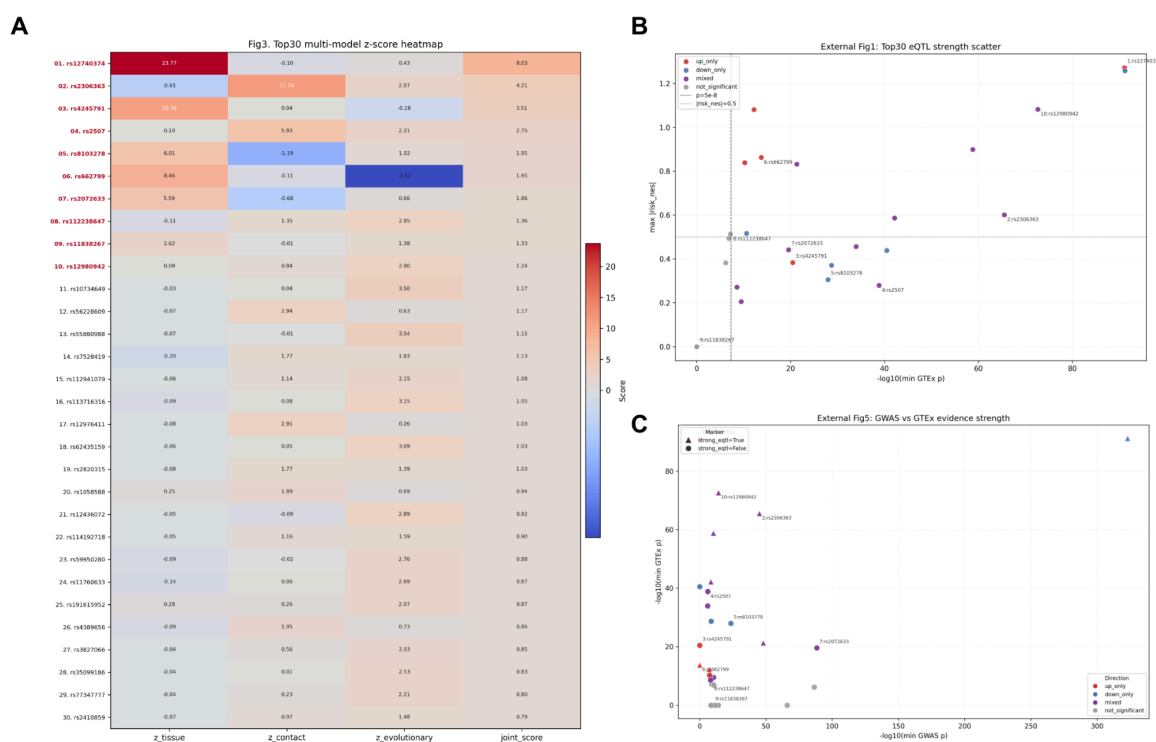
Micro accuracy across three benchmark suites: routing, groundedness, and task success. Bars show the fraction of passed cases for s2f-agent, gpt-4o, and o3-mini; the overall score aggregates all 33 benchmark cases, including 18 routing, 4 groundedness, and 11 task-success cases. s2f-agent achieved 100% accuracy across all suites, whereas gpt-4o reached 22.2% routing, 0% groundedness, 36.4% task success, and 24.2% overall accuracy, and o3-mini reached 27.8% routing, 0% groundedness, 27.3% task success, and 24.2% overall accuracy. Bars are colored by blue, s2f-agent; orange, gpt-4o; green, o3-mini.



Supplementary Figure 2. K562 prediction with pre-trained proxy comparison.

(A) Scatter plots comparing observed fold-change means with post-fine-tuned head predictions across 2,520 held-out chr8 test intervals for ATAC and H3K27ac. Dashed lines indicate the identity line; Pearson correlations were $r = 0.690$ for ATAC and $r = 0.564$ for H3K27ac.

(B-C) Full-test chr8 track comparisons for ATAC (B) and H3K27ac (C). Curves show observed signal, pre-trained proxy output, and post-fine-tuned head prediction across genomic midpoint coordinates. The ATAC proxy used ENCSR921NMD, and the H3K27ac proxy used ENCSR715CCR. Post-head performance was $r = 0.690$, MAE = 0.482 for ATAC and $r = 0.564$, MAE = 2.673 for H3K27ac; corresponding pre-proxy performance was $r = 0.658$, MAE = 5.947 for ATAC and $r = 0.593$, MAE = 2.591 for H3K27ac.



Supplementary Figure 3. Model-score composition and external validation of Top30 CAD variants.

(A) Heatmap of the Top30 CAD-associated noncoding variants ranked by joint score. Columns show AlphaGenome tissue-effect z-score (z_{tissue}), Borzoi contact-perturbation z-score (z_{contact}), GPN evolutionary z-score ($z_{\text{evolutionary}}$), and the final joint_score; rows are ordered by decreasing joint score.

(B) GTEx eQTL evidence strength for the Top30 variants. The x-axis shows $-\log_{10}(\text{min GTEx } p)$, and the y-axis shows the maximum absolute risk-allele-oriented normalized effect size ($\text{max } |\text{risk_nes}|$). Dashed and dotted reference lines mark $p = 5 \times 10^{-8}$ and $|\text{risk_nes}| = 0.5$, respectively; points are colored by the eQTL direction class.

(C) Joint external evidence from GWAS Catalog and GTEx. Each point represents one Top30 variant, plotted by $-\log_{10}(\text{min GWAS } p)$ versus $-\log_{10}(\text{min GTEx } p)$. Colors indicate GTEx direction class, triangles mark strong eQTLs, and Top10 variants are labeled. Overall, 20/30 variants had genome-wide significant GTEx eQTL evidence and 10/30 met the stronger $p < 5 \times 10^{-8}$ and $|\text{risk_nes}| \geq 0.5$ criterion.

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